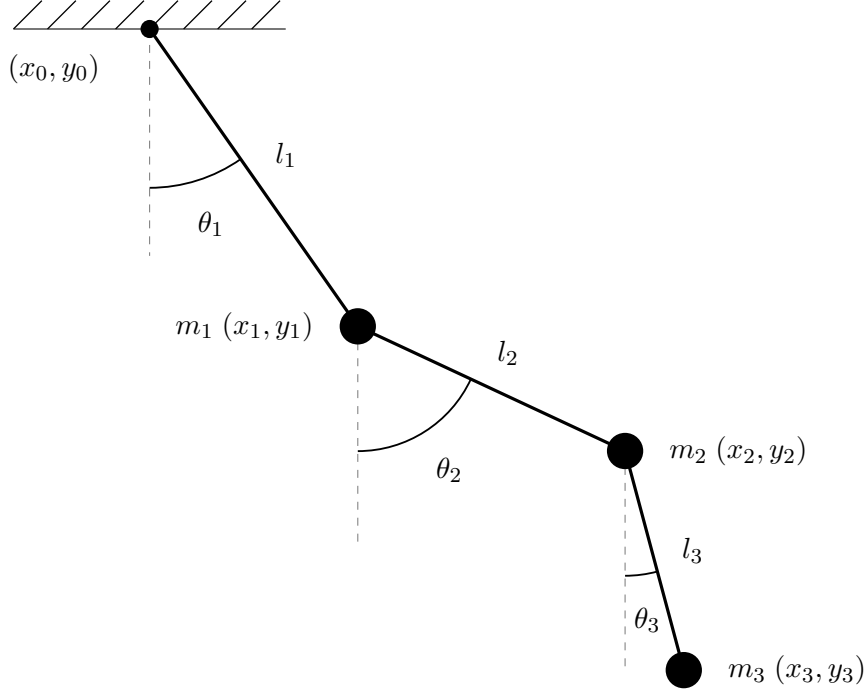


Euler–Lagrange derivation (3-arm pendulum)

Setup

Pivot at the origin. Generalized coordinates: the absolute angles $\theta_1, \theta_2, \theta_3$. Screen convention: $(0,0)$ is the top-left corner, so y increases downward.



Positions

$$\begin{aligned}x_1 &= x_0 + l_1 \sin \theta_1 & y_1 &= y_0 + l_1 \cos \theta_1 \\x_2 &= x_1 + l_2 \sin \theta_2 & y_2 &= y_1 + l_2 \cos \theta_2 \\x_3 &= x_2 + l_3 \sin \theta_3 & y_3 &= y_2 + l_3 \cos \theta_3\end{aligned}$$

Velocities

$$\begin{aligned}\dot{x}_1 &= l_1 \cos \theta_1 \dot{\theta}_1 & \dot{y}_1 &= -l_1 \sin \theta_1 \dot{\theta}_1 \\v_1 &= \sqrt{\dot{x}_1^2 + \dot{y}_1^2} & v_1^2 &= \dot{x}_1^2 + \dot{y}_1^2 \\ \dot{x}_2 &= \dot{x}_1 + l_2 \cos \theta_2 \dot{\theta}_2 & \dot{y}_2 &= \dot{y}_1 - l_2 \sin \theta_2 \dot{\theta}_2 \\v_2 &= \sqrt{\dot{x}_2^2 + \dot{y}_2^2} & v_2^2 &= \dot{x}_2^2 + \dot{y}_2^2 \\ \dot{x}_3 &= \dot{x}_2 + l_3 \cos \theta_3 \dot{\theta}_3 & \dot{y}_3 &= \dot{y}_2 - l_3 \sin \theta_3 \dot{\theta}_3 \\v_3 &= \sqrt{\dot{x}_3^2 + \dot{y}_3^2} & v_3^2 &= \dot{x}_3^2 + \dot{y}_3^2\end{aligned}$$

Kinetic energy

$$T = \frac{1}{2}mv^2 \quad \implies \quad T = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2$$

Potential energy

$$V = mgh, \quad h_1 = -y_1, \quad h_2 = -y_2, \quad h_3 = -y_3$$

$$V = g(m_1h_1 + m_2h_2 + m_3h_3)$$

$$V = -g(m_1y_1 + m_2y_2 + m_3y_3)$$

Lagrangian

$$L = T - V$$

$$L = \frac{1}{2}(m_1v_1^2 + m_2v_2^2 + m_3v_3^2) + g(m_1y_1 + m_2y_2 + m_3y_3)$$

$$L = \frac{1}{2} \left(m_1(\dot{x}_1^2 + \dot{y}_1^2) + m_2(\dot{x}_2^2 + \dot{y}_2^2) + m_3(\dot{x}_3^2 + \dot{y}_3^2) \right) + g(m_1y_1 + m_2y_2 + m_3y_3)$$

Substituting the velocities:

$$\begin{aligned} L = & \frac{1}{2}m_1 \left((l_1 \cos \theta_1 \dot{\theta}_1)^2 + (-l_1 \sin \theta_1 \dot{\theta}_1)^2 \right) \\ & + \frac{1}{2}m_2 \left((\dot{x}_1 + l_2 \cos \theta_2 \dot{\theta}_2)^2 + (\dot{y}_1 - l_2 \sin \theta_2 \dot{\theta}_2)^2 \right) \\ & + \frac{1}{2}m_3 \left((\dot{x}_2 + l_3 \cos \theta_3 \dot{\theta}_3)^2 + (\dot{y}_2 - l_3 \sin \theta_3 \dot{\theta}_3)^2 \right) \\ & + g(m_1y_1 + m_2y_2 + m_3y_3) \end{aligned}$$

$$\begin{aligned} L = & \frac{1}{2}m_1 \left((l_1 \cos \theta_1 \dot{\theta}_1)^2 + (-l_1 \sin \theta_1 \dot{\theta}_1)^2 \right) \\ & + \frac{1}{2}m_2 \left((l_1 \cos \theta_1 \dot{\theta}_1 + l_2 \cos \theta_2 \dot{\theta}_2)^2 + (-l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin \theta_2 \dot{\theta}_2)^2 \right) \\ & + \frac{1}{2}m_3 \left((\dot{x}_1 + l_2 \cos \theta_2 \dot{\theta}_2 + l_3 \cos \theta_3 \dot{\theta}_3)^2 + (\dot{y}_1 - l_2 \sin \theta_2 \dot{\theta}_2 - l_3 \sin \theta_3 \dot{\theta}_3)^2 \right) \\ & + g(m_1(y_0 + l_1 \cos \theta_1) + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) + m_3(y_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \end{aligned}$$

$$\begin{aligned} L = & \frac{1}{2}m_1 \left((l_1 \cos \theta_1 \dot{\theta}_1)^2 + (-l_1 \sin \theta_1 \dot{\theta}_1)^2 \right) \\ & + \frac{1}{2}m_2 \left((l_1 \cos \theta_1 \dot{\theta}_1 + l_2 \cos \theta_2 \dot{\theta}_2)^2 + (-l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin \theta_2 \dot{\theta}_2)^2 \right) \\ & + \frac{1}{2}m_3 \left((l_1 \cos \theta_1 \dot{\theta}_1 + l_2 \cos \theta_2 \dot{\theta}_2 + l_3 \cos \theta_3 \dot{\theta}_3)^2 \right. \\ & \quad \left. + (-l_1 \sin \theta_1 \dot{\theta}_1 - l_2 \sin \theta_2 \dot{\theta}_2 - l_3 \sin \theta_3 \dot{\theta}_3)^2 \right) \\ & + g(m_1(y_0 + l_1 \cos \theta_1) + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\ & \quad + m_3(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \end{aligned}$$

Expanding the squares:

$$(a + b + c)(a + b + c) = a^2 + ab + ac + ab + b^2 + bc + ac + bc + c^2 = a^2 + 2ab + b^2 + 2bc + c^2 + 2ac$$

$$\begin{aligned} L = & \frac{1}{2}m_1 \left(l_1^2 \cos^2 \theta_1 \dot{\theta}_1^2 + l_1^2 \sin^2 \theta_1 \dot{\theta}_1^2 \right) \\ & + \frac{1}{2}m_2 (l_1^2 \cos^2 \theta_1 \dot{\theta}_1^2 + 2 \cdot l_1 \cos \theta_1 \dot{\theta}_1 l_2 \cos \theta_2 \dot{\theta}_2 + l_2^2 \cos^2 \theta_2 \dot{\theta}_2^2 \\ & \quad + l_1^2 \sin^2 \theta_1 \dot{\theta}_1^2 + 2 \cdot l_1 \sin \theta_1 \dot{\theta}_1 l_2 \sin \theta_2 \dot{\theta}_2 + l_2^2 \sin^2 \theta_2 \dot{\theta}_2^2) \\ & + \frac{1}{2}m_3 (l_1^2 \cos^2 \theta_1 \dot{\theta}_1^2 + l_2^2 \cos^2 \theta_2 \dot{\theta}_2^2 + l_3^2 \cos^2 \theta_3 \dot{\theta}_3^2 \\ & \quad + 2 \cdot l_1 \cos \theta_1 \dot{\theta}_1 l_2 \cos \theta_2 \dot{\theta}_2 + 2 \cdot l_1 \cos \theta_1 \dot{\theta}_1 l_3 \cos \theta_3 \dot{\theta}_3 \\ & \quad + 2 \cdot l_2 \cos \theta_2 \dot{\theta}_2 l_3 \cos \theta_3 \dot{\theta}_3 \\ & \quad + l_1^2 \sin^2 \theta_1 \dot{\theta}_1^2 + l_2^2 \sin^2 \theta_2 \dot{\theta}_2^2 + l_3^2 \sin^2 \theta_3 \dot{\theta}_3^2 \\ & \quad + 2 \cdot l_1 \sin \theta_1 \dot{\theta}_1 l_2 \sin \theta_2 \dot{\theta}_2 + 2 \cdot l_1 \sin \theta_1 \dot{\theta}_1 l_3 \sin \theta_3 \dot{\theta}_3 \\ & \quad + 2 \cdot l_2 \sin \theta_2 \dot{\theta}_2 l_3 \sin \theta_3 \dot{\theta}_3) \\ & + g(m_1(y_0 + l_1 \cos \theta_1) \\ & \quad + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\ & \quad + m_3(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \end{aligned}$$

$$\begin{aligned} L = & \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) \\ & + \frac{1}{2}m_2 (l_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) + l_2^2 \dot{\theta}_2^2 (\cos^2 \theta_2 + \sin^2 \theta_2) \\ & \quad + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2)) \\ & + \frac{1}{2}m_3 (l_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) + l_2^2 \dot{\theta}_2^2 (\cos^2 \theta_2 + \sin^2 \theta_2) + l_3^2 \dot{\theta}_3^2 (\cos^2 \theta_3 + \sin^2 \theta_3) \\ & \quad + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) \\ & \quad + 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 (\cos \theta_1 \cos \theta_3 + \sin \theta_1 \sin \theta_3) \\ & \quad + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 (\cos \theta_2 \cos \theta_3 + \sin \theta_2 \sin \theta_3)) \\ & + g(m_1(y_0 + l_1 \cos \theta_1) \\ & \quad + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\ & \quad + m_3(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \end{aligned}$$

Using $\cos^2 + \sin^2 = 1$ and $\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 = \cos(\theta_1 - \theta_2)$:

$$\begin{aligned} L = & \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 \\ & + \frac{1}{2}m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\ & + \frac{1}{2}m_3 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + l_3^2 \dot{\theta}_3^2 \\ & \quad + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\ & \quad + 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \cos(\theta_1 - \theta_3) \\ & \quad + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \cos(\theta_2 - \theta_3)) \\ & + g(m_1(y_0 + l_1 \cos \theta_1) \\ & \quad + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\ & \quad + m_3(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \end{aligned}$$

Euler–Lagrange

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

For θ_1

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = 0$$

$$\begin{aligned} \frac{\partial L}{\partial \theta_1} = \frac{\partial}{\partial \theta_1} & \left[\frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 \right. \\ & + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\ & + \frac{1}{2} m_3 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + l_3^2 \dot{\theta}_3^2 \\ & \quad + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\ & \quad + 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \cos(\theta_1 - \theta_3) \\ & \quad + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \cos(\theta_2 - \theta_3)) \\ & + g(m_1(y_0 + l_1 \cos \theta_1) \\ & \quad + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\ & \quad \left. + m_3(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \right] \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial \theta_1} = \frac{1}{2} m_2 & (-2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)) \\ & + \frac{1}{2} m_3 (-2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\ & \quad - 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \sin(\theta_1 - \theta_3)) \\ & + g(-m_1(l_1 \sin \theta_1) - m_2(l_1 \sin \theta_1) - m_3(l_1 \sin \theta_1)) \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial \theta_1} = & -m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\ & - m_3 (l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\ & \quad + l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \sin(\theta_1 - \theta_3)) \\ & - g l_1 (m_1 + m_2 + m_3) \sin \theta_1 \end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial \dot{\theta}_1} = \frac{\partial}{\partial \dot{\theta}_1} \left[\frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 \right. \\ + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\ + \frac{1}{2} m_3 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + l_3^2 \dot{\theta}_3^2 \\ + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\ + 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \cos(\theta_1 - \theta_3) \\ + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \cos(\theta_2 - \theta_3)) \\ + g(m_1(y_0 + l_1 \cos \theta_1) \\ + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\ \left. + m_3(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \right]\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial \dot{\theta}_1} = m_1 l_1^2 \dot{\theta}_1 \\ + \frac{1}{2} m_2 (2 \cdot l_1^2 \dot{\theta}_1 + 2 \cdot l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\ + \frac{1}{2} m_3 (2 \cdot l_1^2 \dot{\theta}_1 + 2 \cdot l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + 2 \cdot l_1 l_3 \dot{\theta}_3 \cos(\theta_1 - \theta_3))\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial \dot{\theta}_1} = m_1 l_1^2 \dot{\theta}_1 \\ + m_2 (l_1^2 \dot{\theta}_1 + l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\ + m_3 (l_1^2 \dot{\theta}_1 + l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + l_1 l_3 \dot{\theta}_3 \cos(\theta_1 - \theta_3))\end{aligned}$$

$$\begin{aligned}\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = \frac{d}{dt} \left[m_1 l_1^2 \dot{\theta}_1 \right. \\ + m_2 (l_1^2 \dot{\theta}_1 + l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\ \left. + m_3 (l_1^2 \dot{\theta}_1 + l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + l_1 l_3 \dot{\theta}_3 \cos(\theta_1 - \theta_3)) \right]\end{aligned}$$

$$\begin{aligned}\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = m_1 l_1^2 \ddot{\theta}_1 \\ + m_2 (l_1^2 \ddot{\theta}_1 + l_1 l_2 (\ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2))) \\ + m_3 (l_1^2 \ddot{\theta}_1 \\ + l_1 l_2 (\ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2)) \\ + l_1 l_3 (\ddot{\theta}_3 \cos(\theta_1 - \theta_3) - \dot{\theta}_3 \sin(\theta_1 - \theta_3) (\dot{\theta}_1 - \dot{\theta}_3)))\end{aligned}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = 0$$

$$\begin{aligned}
0 = & m_1 l_1^2 \ddot{\theta}_1 \\
& + m_2 \left(l_1^2 \ddot{\theta}_1 + l_1 l_2 [\ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \dot{\theta}_2 \sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2)] \right) \\
& + m_3 \left(l_1^2 \ddot{\theta}_1 \right. \\
& \quad + l_1 l_2 [\ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \dot{\theta}_2 \sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2)] \\
& \quad + l_1 l_3 [\ddot{\theta}_3 \cos(\theta_1 - \theta_3) - \dot{\theta}_3 \sin(\theta_1 - \theta_3)(\dot{\theta}_1 - \dot{\theta}_3)] \Big) \\
& - \left(-m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \right. \\
& \quad - m_3 [l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& \quad \quad + l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \sin(\theta_1 - \theta_3)] \\
& \quad \left. - gl_1(m_1 + m_2 + m_3) \sin \theta_1 \right)
\end{aligned}$$

$$\begin{aligned}
0 = & m_1 l_1^2 \ddot{\theta}_1 \\
& + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) - m_2 l_1 l_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2) \\
& + m_3 l_1^2 \ddot{\theta}_1 \\
& + m_3 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) - m_3 l_1 l_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2) \\
& + m_3 l_1 l_3 \ddot{\theta}_3 \cos(\theta_1 - \theta_3) - m_3 l_1 l_3 \dot{\theta}_3 \sin(\theta_1 - \theta_3)(\dot{\theta}_1 - \dot{\theta}_3) \\
& + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \sin(\theta_1 - \theta_3) \\
& + gl_1(m_1 + m_2 + m_3) \sin \theta_1
\end{aligned}$$

$$\begin{aligned}
0 = & m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_3 l_1^2 \ddot{\theta}_1 \\
& + m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& - m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_2 l_1 l_2 \dot{\theta}_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& - m_3 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_2 \dot{\theta}_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_3 \ddot{\theta}_3 \cos(\theta_1 - \theta_3) \\
& - m_3 l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \sin(\theta_1 - \theta_3) \\
& + m_3 l_1 l_3 \dot{\theta}_3 \dot{\theta}_3 \sin(\theta_1 - \theta_3) \\
& + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \\
& + m_3 l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \sin(\theta_1 - \theta_3) \\
& + gl_1(m_1 + m_2 + m_3) \sin \theta_1
\end{aligned}$$

$$\begin{aligned}
0 = & m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_3 l_1^2 \ddot{\theta}_1 \\
& + m_2 \left[l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \right. \\
& \quad \left. + l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \right] \\
& + m_3 \left[l_1 l_3 \ddot{\theta}_3 \cos(\theta_1 - \theta_3) \right. \\
& \quad + l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& \quad + l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\
& \quad \left. + l_1 l_3 \dot{\theta}_3^2 \sin(\theta_1 - \theta_3) \right] \\
& + g l_1 (m_1 + m_2 + m_3) \sin \theta_1
\end{aligned}$$

Solution for θ_1

$$\begin{aligned}
& (m_1 + m_2 + m_3) l_1^2 \ddot{\theta}_1 + (m_2 + m_3) l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& + m_3 l_1 l_3 \ddot{\theta}_3 \cos(\theta_1 - \theta_3) = (-m_2 - m_3) l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\
& - m_3 l_1 l_3 \dot{\theta}_3^2 \sin(\theta_1 - \theta_3) - g l_1 (m_1 + m_2 + m_3) \sin \theta_1
\end{aligned}$$

For θ_2

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) - \frac{\partial L}{\partial \theta_2} = 0$$

$$\begin{aligned}
\frac{\partial L}{\partial \theta_2} = & \frac{\partial}{\partial \theta_2} \left[\frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 \right. \\
& + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\
& + \frac{1}{2} m_3 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + l_3^2 \dot{\theta}_3^2 \\
& \quad + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& \quad + 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \cos(\theta_1 - \theta_3) \\
& \quad \left. + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \cos(\theta_2 - \theta_3) \right) \\
& + g (m_1 (y_0 + l_1 \cos \theta_1) \\
& \quad + m_2 (y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\
& \quad \left. + m_3 (y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3) \right) \Big]
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial \theta_2} = & \frac{1}{2} m_2 \left(2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2) \cdot (-1)) \right) \\
& + \frac{1}{2} m_3 \left(2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2) \cdot (-1)) \right. \\
& \quad \left. + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 (-\sin(\theta_2 - \theta_3) \cdot (-1)) \right) \\
& + g \left(m_2 (l_2 (-\sin \theta_2)) + m_3 l_2 (-\sin \theta_2) \right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial \dot{\theta}_2} = \frac{\partial}{\partial \dot{\theta}_2} \bigg[& \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 \\
& + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \\
& + \frac{1}{2} m_3 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + l_3^2 \dot{\theta}_3^2 \\
& \quad + 2 \cdot l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\
& \quad + 2 \cdot l_1 l_3 \dot{\theta}_1 \dot{\theta}_3 \cos(\theta_1 - \theta_3) \\
& \quad + 2 \cdot l_2 l_3 \dot{\theta}_2 \dot{\theta}_3 \cos(\theta_2 - \theta_3)) \\
& + g(m_1(y_0 + l_1 \cos \theta_1) \\
& \quad + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2) \\
& \quad + m_3(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3)) \bigg]
\end{aligned}$$